

SNAPFRAME · DESIGN PRECIS

SnapFrame design precis

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
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Source	docs/02-concept.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.8	2026-10-02	Amish Chadha	Decisions of 2026-10-02 carried into the design: folding step, three tarpaulins with the cutting plan, long rear anchors, filled nylon nodes, frame analysis (rating 19.5 m/s), two tube bundles; mass, packages and cost from SNF-CAL-001 v0.6
0.7	2026-10-02	Amish Chadha	Gable, packages, polymer, anchors, posts, offcut and partner decided by Amish on 2026-10-02 (SNF-DEC-001)
0.6	2026-10-01	Amish Chadha	Design for construction (SNF-DDR-003, Draft) applied; erection order, joints and cable fixings updated; cost against the value-engineering target
0.5	2026-09-26	Amish Chadha	Budget approved by Amish (\$445, SNF-DDR-002); cost figure restated against it
0.4	2026-09-25	Amish Chadha	Recommendations accepted by Amish (DDR-002)
0.3	2026-09-25	Amish Chadha	Apply SNF-DDR-001 (1 in rafters, frame kit budget, decided design)

choices); numbers from SNF-CAL-001; parametric model and SNF-DWG-001

0.2	2026-09-25	Amish Chadha	Populate to TRL 2 (architecture, sizes, first-order numbers, design choices, safety, media)
0.1	2026-09-24	Amish Chadha	Initial scaffold

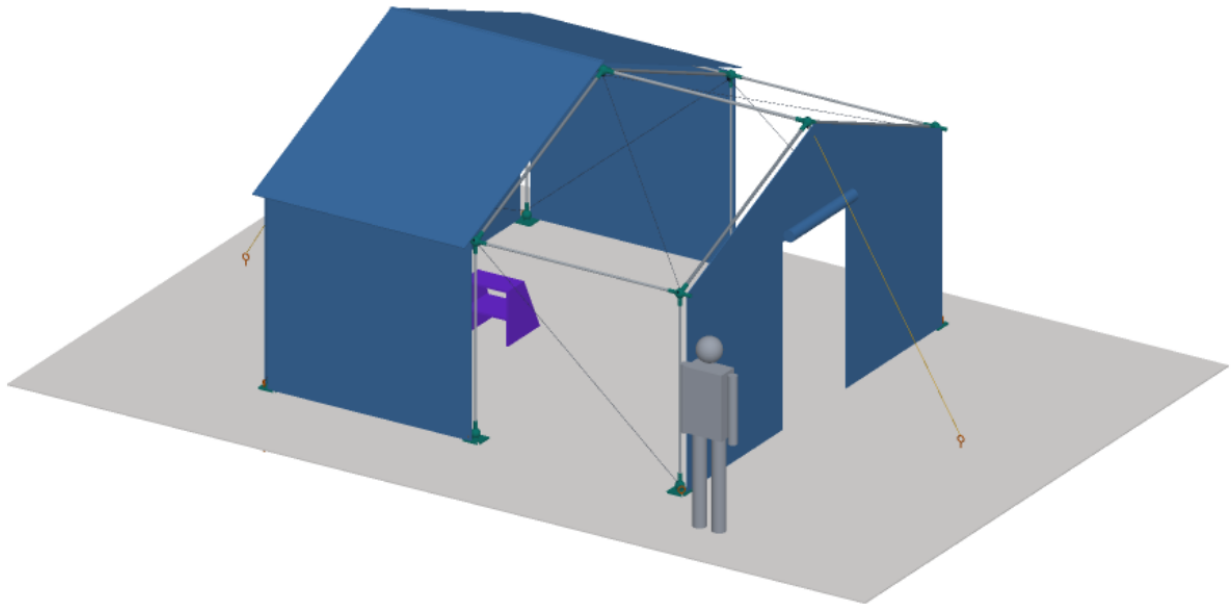
SnapFrame design precis

SnapFrame is a gable-roof shelter frame of straight EMT conduit joined by printed nodes: 1 in EMT rafters and ridge tubes, and 3/4 in EMT posts and eave tubes. Each tube end carries a spring button that snaps into a hole in the node socket, so two adults can put the frame up by hand. The reference size M covers 4.0 x 4.0 m (16 m²) with a 2.6 m ridge and takes three standard 4 x 6 m relief tarpaulins from agency stock as its skin (two for the roof, side walls and rear gable, the third cut for the front gable with a door flap). The TRL 3 calculation note SNF-CAL-001 v0.6 puts the constructable frame kit (SNF-DDR-003) at about 46.2 kg and \$701, \$256 over the \$445 value-engineering target (the nodes are in filled PA12-class nylon), and rates the frame at about 19.5 m/s (70 km/h, 44 mph), just short of the 20 m/s target, with the 3/4 in posts governing in the frame analysis.

SnapFrame

github.com/BoujeeEnjinia1701/snapframe

CONCEPT, NOT FOR FABRICATION



Seen from the front right and above, 24 deg elevation. Grey: ground patch, 1.75 m person for scale

Figure 1. SnapFrame size M with a 1.75 m person for scale. The tarpaulin skin is shown on the rear bay and the front gable (door flap rolled up), so the frame, nodes and bracing stay visible between them. Generated from the parametric model `cad/src/model.py`.

■ How it works

1. **Unpack.** The frame kit arrives as two strapped bundles of straight tubes (18 in all, four types, color-coded at the ends, snap buttons fitted) and a bag with 15 nodes (three families, four printed variants, cable bolts fitted to the

eave and ridge nodes), 10 brace cables, 8 screw anchors (two of them longer), 2 guy lines, a folding step, hitch pins and caps. Three tarpaulins come from agency stock.

2. **Mark out.** Six pegs mark the feet with a knotted layout cord (4.0 x 4.0 m, diagonals equal).
3. **Build the frames.** Each of the three gable frames is two posts, two rafters, two eave nodes, one ridge node and two foot nodes, assembled flat on the ground. Every joint closes with a click as the spring button finds its hole; hitch pins go in at the feet and the eave post sockets.
4. **Stand and join the frames.** The rear frame is walked up and stood on its marks. The ridge and eave tubes go into it, and the middle frame, stood 70 mm short of its marks, slides 65 mm on its feet onto the three tube ends; the front bay follows the same way (SNF-DDR-003 C7). A folding step in the kit reaches the ridge sockets at 2.6 m (decided 2026-10-02; guiding from the ground is timed at TRL 4). Then the frame is squared, and a screw anchor is turned in by hand through a slot in each foot plate, using a spare tube through its eye as a lever, until the eye sits down on the plate.
5. **Brace.** Brace cables with hand tensioners go into the rear gable, one bay of each side wall and one diagonal per bay in each roof plane. Their snap hooks clip to the anchor eyes at the feet and to the steel ring on each eave and ridge node's cable bolt, and they are pulled snug. Guy lines run from the rings of the two end ridge nodes to anchors 1.5 m beyond each gable.
6. **Skin.** One tarpaulin goes over the ridge as the roof; the second is cut into three strips that close both side walls and the lower rear gable. Part of a third tarpaulin from agency stock, cut to include a door flap, closes the front gable, and the gable tops come from its offcut (R7, decided 2026-10-02; cutting plan in the build plan, Figure 18). The blank socket on each corner eave node takes a push-in cap. Tarpaulins are tied through their eyelets to the tubes, never to the nodes.
7. **Strike and reuse.** Pressing each button in its finger recess releases its joint. Tubes and nodes go back in the bundle and bag, or the frame stays and carries better cladding later.

■ Main components

Table 1. Main components. Numbers match `bom/bom.csv`, Figure 2 and drawing SNF-DWG-001.

#	Component	Choice	Notes
1	Posts (6)	3/4 in EMT, 1.650 m	One 3.05 m (10 ft) stick each
2	Rafters (6)	1 in EMT, 2.064 m	SNF-DDR-001 D2; second socket bore on eave and ridge nodes
3	Ridge tubes (2)	1 in EMT, 1.910 m	SNF-DDR-002 D8 (were 3/4 in and governed the wind rating)
4	Eave tubes (4)	3/4 in EMT, 1.910 m	Same length as item 3, smaller tube; different end color for the guide
5	Foot nodes (6)	Printed in filled PA12-class nylon, one socket, 170 x 170 x 12 mm plate with two anchor slots at 45°	About 460 g each
6	Eave nodes (6)	Printed, four sockets and a cable bolt hole, one variant	At the corners the socket past the gable stays blank and capped (SNF-DDR-002 D9); about 287 g
7	Ridge nodes (3)	Printed, three or four sockets, all 30.1 mm bore, and a cable bolt hole	About 277 to 304 g
8	Brace cables (10)	4 mm galvanized wire rope, thimble loop, hand tensioner, snap hook at each end	29.5 m eye to eye
9	Screw ground anchors (6)	380 mm galvanized screw anchor with eye	Four at the middle and front feet, two for guys
10	Guy lines (2)	6 mm polyester rope with slide tensioner	From end ridge nodes
11	Skin	Three 4 x 6 m reinforced polyethylene tarpaulins	Agency stock, outside the kit budget (SNF-DDR-001 D1); the third closes the front gable with a door flap
12	Snap buttons, hitch pins and caps	36 spring buttons (one per tube end), 12 hitch pins on lanyards, 4 socket caps	Hitch pins at the 12 tension joints (SNF-DDR-001 D4)
13	Bundle straps and bag	Four cam straps, two for each tube bundle; one duffel bag	
14	Cable bolt sets (9)	Stainless M10 bolt through each eave and ridge node, with spacer, washers, nyloc nut and a welded 6 mm ring	The cables and guy lines clip to the ring (SNF-DDR-003 C3)
15	Folding step (1)	Two-tread steel or aluminum step, top tread 0.50 m, about 1.5 kg	Reaches the ridge sockets at 2.6 m (SNF-DDR-003 A1, decided 2026-10-02)
16	Long screw anchors (2)	560 mm galvanized screw anchor with the same eye, rated 1.5 kN	One at each rear corner foot (R8, raised 2026-10-02)

Node concept

A node is a spherical core (84 mm diameter) with one socket per member. Each socket is 110 mm long from the node center; the tube end stops 45 mm from the center, so it engages 65 mm. The bore is the tube outside diameter plus 0.6 mm (24.0 mm for 3/4 in, 30.1 mm for 1 in) and the socket wall is 5 mm. One eave node serves all six eave

positions; it is symmetric about its own frame line, so it is not handed, and at the corners its outward socket stays empty under a cap. A 6 mm spring button hole sits 40 mm from the tube end, in a 16 mm finger recess so the button can be pressed by hand; at the feet and eave post sockets an 8 mm hitch pin hole sits 15 mm from the tube end, clear of the button's spring inside the tube (SNF-DDR-003 C1, C2). Each socket mouth has a lead-in chamfer. Each eave and ridge node carries a through-bolted M10 cable bolt with a steel ring, and the foot plate has two anchor slots, so the cables never load the print in bending (SNF-DDR-003 C3 to C6). Socket axes are computed from the node positions, so a new width, length or pitch produces a new set of printable nodes. The prototype prints the nodes, and the printed node is intended to become a pattern for sand-cast aluminum later (SNF-DDR-001 D3); castability is not checked at TRL 3.

■ Sizes

Table 2. Size family generated from the same parameters (SNF-DDR-001 D6). The model exports the node set of every size.

Size	Floor	Bays	Eave and ridge	People at 3.5 m ² each	Members	Nodes
S	3.0 x 3.0 m (9 m ²)	2 x 1.5 m	1.8 m, 2.4 m	2	18	15 (new angles)
M (reference)	4.0 x 4.0 m (16 m ²)	2 x 2.0 m	1.8 m, 2.6 m	4	18	15
L	4.0 x 6.0 m (24 m ²)	3 x 2.0 m	1.8 m, 2.6 m	6	25	20 (same types as M)

■ Key numbers

All values are from SNF-CAL-001, which lists its assumptions. They are calculations on paper, not measurements.

Table 3. Geometry, mass, cost and wind, size M.

Quantity	Value	Requirement
Floor area	16.0 m ²	R1 met
Headroom 2.0 m or more	73 % of floor	R2 met
Tube length in the kit	33.7 m (54.9 m bought, 39 % offcut)	
Frame kit mass	46.2 kg (102 lb); 59.9 kg with three tarpaulins	R11 met: frame kit alone counted since 2026-10-02 (tarpaulins issued separately)
Packages	Two tube bundles, 16.7 and 12.3 kg, 0.017 and 0.013 m ³ ; bag 17.2 kg, 0.075 m ³ (30.9 kg with three tarpaulins)	R5 met (three packages, restated 2026-10-02)
Erection time	About 46 min for two adults (estimate)	R4 not verifiable at TRL 3
Skin area	51.4 m ² for full enclosure; the cutting plan uses 59.6 m ² of the 72 m ² in three tarpaulins	R7 met, cutting plan drawn (third tarpaulin decided 2026-10-02)
Frame kit cost	About \$701; value-engineering target \$445	R10: \$256 over the value-engineering target (nodes \$316 in filled nylon, \$120 in ASA)
Dynamic pressure at 20 m/s	245 Pa	
Rafter (1 in) at 20 m/s	142 N·m, 166 MPa, factor 1.66	Meets 1.5
Ridge tube (1 in) at 20 m/s	106 N·m, 124 MPa, factor 2.22	Meets 1.5 (1.18 with 3/4 in)
Post (3/4 in) at 20 m/s	86 N·m and 369 N axial, 192 MPa, factor 1.43 (frame analysis; 1.46 bending only)	R6 not met
Wind rating	19.5 m/s (70 km/h, 44 mph); 14.7 m/s with the door flap open and wind into it	R6 not met (posts govern)
Worst foot anchor uplift	1.28 kN at a rear corner, before pretension	R8 at risk (1.5 kN target at the rear corner feet since 2026-10-02)
Snow	Not rated; 0.5 kPa would put 416 MPa in a 1 in rafter	Out of scope

The TRL 2 estimates (factor 1.70 for posts, 1.26 for 3/4 in rafters, about 18 m/s) spread half of each panel uniformly over the frames. SNF-CAL-001 uses 45° tributary lines, which peak at midspan and give higher moments. On that basis, a 3/4 in rafter would reach only 0.89.

SnapFrame: exploded view (skin, item 11, not shown)

github.com/BoujeeEnjinia1701/snapframe

CONCEPT, NOT FOR FABRICATION

- 1 EMT post, 3/4 in, 1.65 m
- 2 EMT rafter, 1 in, 2.06 m
- 3 EMT ridge tube, 1 in, 1.91 m
- 4 EMT eave tube, 3/4 in, 1.91 m
- 5 Foot node with two anchor slots
- 6 Eave node
- 7 Ridge node
- 8 Brace cable with hand tensioner
- 9 Screw ground anchor
- 10 Guy line
- 14 Cable bolt set and ring
- 15 Folding step
- 16 Long screw anchor, rear corner (560 mm)

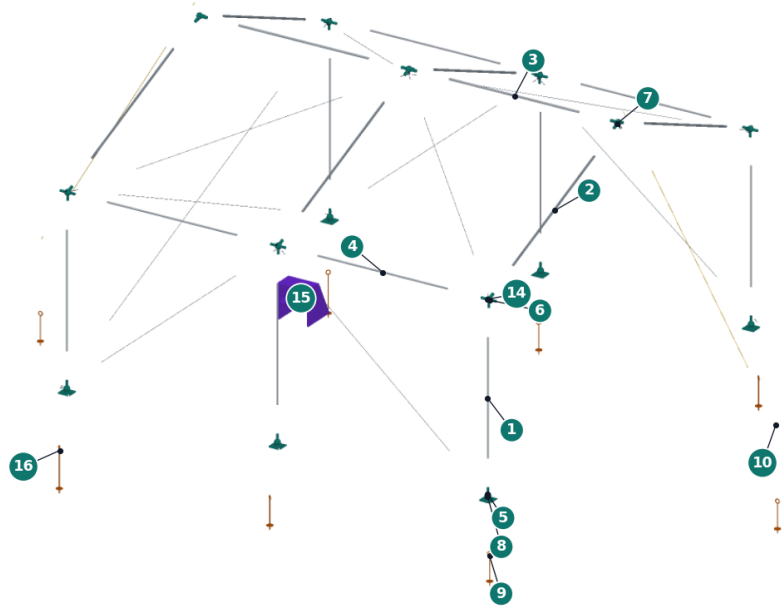


Figure 2. Exploded view with BOM numbers. The skin (item 11) is left out so the frame parts stay visible; items 12 and 13 are not modeled; item 15 is the folding step and item 16 the long rear anchors.

■ Design choices

Decided by Amish on 2026-09-25 (SNF-DDR-001):

- **Tube size (D2).** 1 in EMT rafters, with the wind rating stated on the kit.
- **Budget scope (D1).** The budget covers the frame kit; Amish set it at \$445 on 2026-09-26 (SNF-DDR-002) and on 2026-10-01 made it a value-engineering target, not a limit. Tarpaulins come from agency stock and are costed separately.
- **Node process (D3).** Print for the first prototype, design every node to be castable from the printed pattern.
- **Joint locking (D4).** Spring buttons at every tube end, hitch pins at the 12 tension joints.
- **Frame form (D5).** Gable roof with pinned nodes and cable bracing, not a dome, barrel vault or rigid nodes.
- **Sizes (D6)** S, M and L, with M as the reference, and **scope (D7)** without snow or cyclone rating in the first release.

Decided by Amish on 2026-09-25, going with the recommendations (SNF-DDR-002):

- **Ridge tubes (D8).** 1 in EMT ridge tubes as well, which lifts the rating from 17.8 to 19.7 m/s by the member check, and to 19.5 m/s once the frame analysis (run 2026-10-02) adds axial force. The posts stay 3/4 in; 1 in posts would give 20.3 m/s for about \$30 more.
- **One eave node (D9).** The four-socket eave node is used at the corners with one blank, capped socket, so there are no handed nodes and R12 is met.
- **Wind rating on the kit (D2 with D8).** Until R6 is met the kit label states 19.5 m/s (70 km/h) and about 14.7 m/s with the door flap open and wind into it.

Decided by Amish on 2026-10-02 (SNF-DEC-001): the design for construction changes of SNF-DDR-003 are accepted; the 39 % tube offcut is accepted for the prototype, with 1.50 m posts or 6 m metric stock chosen once the first region's tube source is known; the IFRC Shelter Research Unit is the first candidate to approach for a technical review and Field Ready for printing; part of a third tarpaulin closes the front gable (R7); the frame analysis is run now (done in SNF-CAL-001 v0.6, posts stay 3/4 in) (R6); a folding step joins the kit for the first prototype; the tubes go in two bundles, R5 is three packages and R11 counts the frame kit alone; the nodes are printed in glass- or carbon-filled PA12-class nylon (R9); and the rear corner anchors are rated 1.5 kN (R8). Cost is followed against the value-engineering target in the register rather than as a decision.

■ Safety

Safety: SnapFrame is an emergency shelter frame, not a storm refuge. By calculation (SNF-CAL-001) it stays elastic with a factor of 1.5 only up to gusts of about 19.5 m/s (70 km/h, 44 mph), and about 14.7 m/s (53 km/h) when wind blows into the open door. Above that, or when a storm warning is issued, occupants should drop the skin, leave the shelter and follow local evacuation guidance. A collapsing steel frame can injure people inside.

Safety: The frame is not rated for snow. Snow or ponding water on the roof can overload the rafters within hours. Clear the roof, or leave the shelter, whenever snow or standing water collects.

Safety: Polyethylene tarpaulins burn quickly and drip. Keep cooking fires, candles, kerosene lamps and stoves outside the shelter or behind a non-combustible screen, keep at least 2 m between shelters where the site allows, and keep the doorway clear. A fire-retardant tarpaulin option is an open requirement (R13).

- **Erection.** Walking a frame up takes two people. Keep hands clear of sockets as tubes drop in (pinch points), and never erect or strike the frame in strong wind, when the tarpaulin acts as a sail.
- **Tube ends.** Cut EMT ends are sharp. Every tube is deburred during kit preparation, and ends sit inside sockets once assembled.
- **Guy lines and anchors.** Guy lines are a trip hazard at night; mark them with reflective tape. Screw anchors must be clear of buried cables and pipes.
- **Electrical.** The frame is conductive steel. Any lighting inside must be low-voltage DC (for example solar lanterns); mains cables must not be run on or near the frame.
- **Lightning.** A steel frame gives no reliable lightning protection. Leave it during thunderstorms where a solid building is available.

■ Open questions

- R6: with 1 in ridge tubes the 3/4 in posts (factor 1.43 in the frame analysis) still fall short at 20 m/s. Whether a real socket gives the knee fixity that would lift the factor to 1.49 or more is a TRL 4 test; 1 in posts remain an option (SNF-CAL-001 v0.6).
- R5 and R11: decided 2026-10-02, two tube bundles of 16.7 and 12.3 kg (three packages) and the carried mass counted on the frame kit alone, 46.2 kg (SNF-DEC-001).
- Confirm EMT yield strength and dimensions from supplier data for the first region, and whether metric thin-wall tube of about 25 mm is stocked where EMT is not.

- Node polymer: glass- or carbon-filled PA12-class nylon chosen on 2026-10-02; printed coupons to be tested for pin bearing at 70 °C, and its strength under UV over two years confirmed (R9).
- Anchor holding capacity in sand, clay and gravel, and the rating of the hand cable tensioners (R8).
- Fire-retardant tarpaulin sources and cost (R13).
- Picture-only erection guide: test with users who have not seen the kit.
- Validate floor area, headroom, erection time, wind exposure and price with users through a local partner.

Drawings and media: [prototype build plan SNF-BLD-001](#), [design decisions register SNF-DEC-001](#), [general arrangement SNF-DWG-001](#), [blueprint sheet](#), [interactive 3D model](#).